

BENZENE

SUBTOPICS

⇒ Diazotisation

⇒ OMSGAP

⇒ Different names of Benzene

⇒ Methods of Preparation of benzene.

⇒ Isotopic Effect.

when $\frac{K_H}{K_D} = 7$

Sulphonation
Iodination.

1. From alkenes
2. Alkynes
3. Aryl halides
4. Benzoic acid

• EDG $\propto$ ESR

• EWG $\propto$ NSR

$$\text{Rate} = k[\text{Benzene}][E^+]$$

⇒ Nitration (HNO_3) : NO_2^+

⇒ Sulphonation (H_2SO_4) : SO_3H^+

⇒ Desulphonation

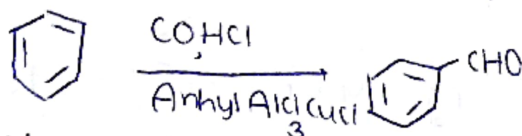
⇒ Halogenation

⇒ F.C Alkylation, Acylation

⇒ Gattermann Rxn.

*
*
*

Benzene $\rightarrow$ Benzaldehyde



⇒ O Cresol, Cumene

⇒ Catechol, Resorcinol

⇒ Substitution at Benzylic Position

⇒ Side chain Oxidation ($\text{KMnO}_4/\text{OH}^-$)

⇒ Arenium ion.

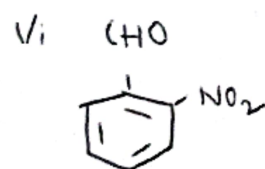
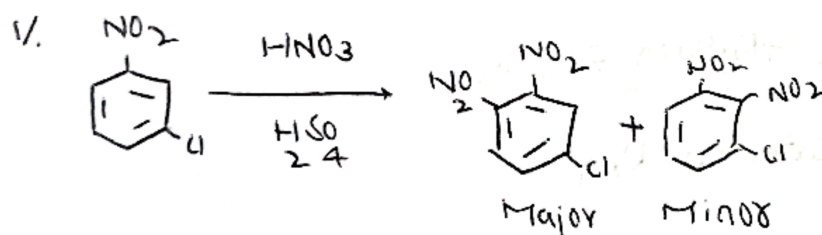
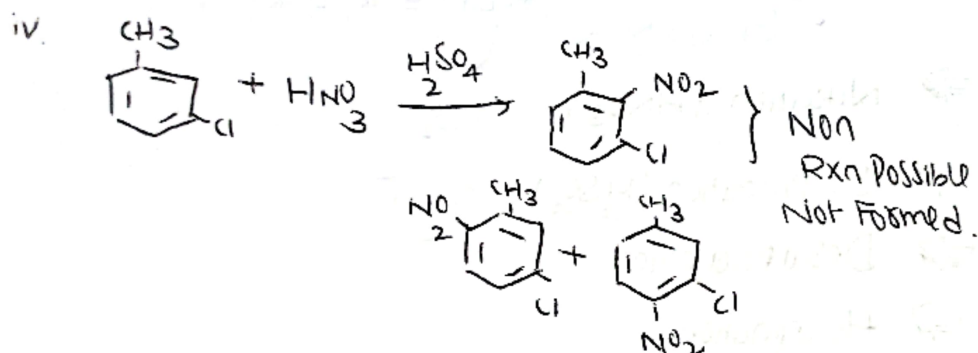
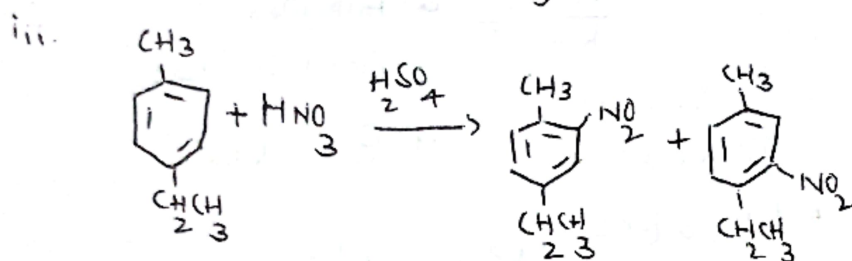
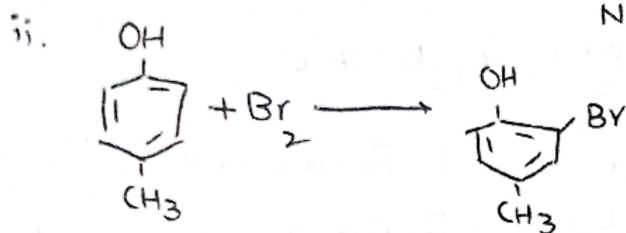
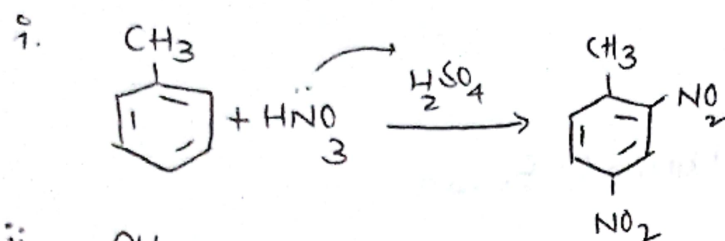
$\text{K}_2\text{Cr}_2\text{O}_7/\text{H}_2\text{SO}_4$ or CrO_3

H_2CrO_4 (or) CrO_4^{2-} with HNO_3

Not applies
when NO
 αH is
Present

CONCEPTS :

JEE Adv Level : (Major, Minor)



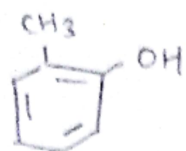
1,4 DW $\rightarrow$ E⁺ Ortho to D

1,4 DD $\rightarrow$ E⁺ Ortho to more DG

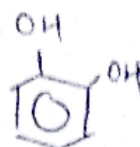
1,3 DD $\rightarrow$ E⁺ at 4 not possible, Ortho is

if there is 2 meta Directing Group Stable the more deactivating group decides Orientation.

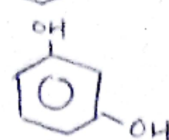
• 2-Methyl Phenol (O-Cresol)



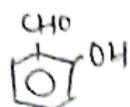
• Catechol (1,2-Dihydroxy benzene)



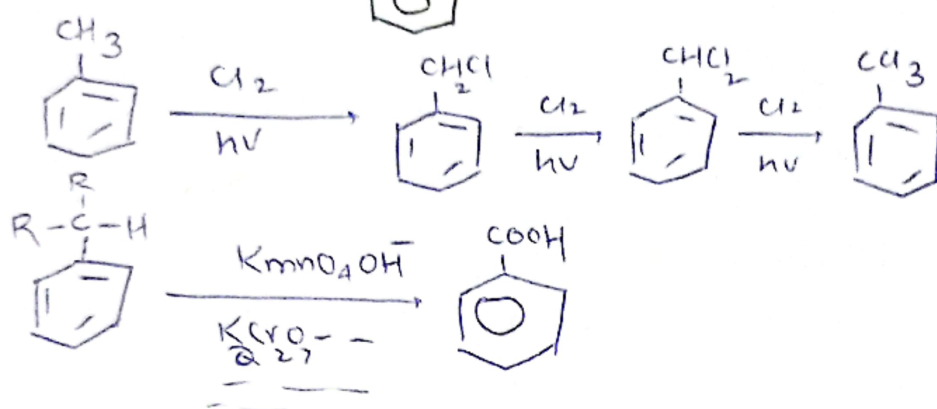
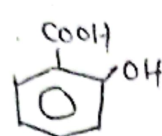
Resorcinol (1,3-Dihydroxy benzene)



Salicylaldehyde



O-Salicylic Acid

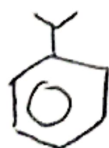


1) Arenium ion:

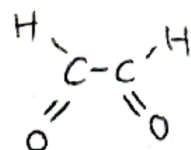
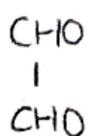
Carbocation formed by electrophilic attack on an aromatic ring.

*
*
*

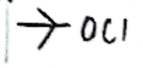
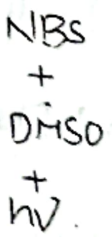
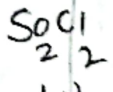
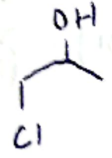
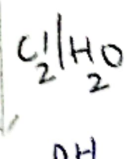
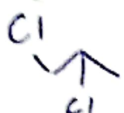
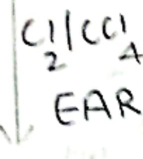
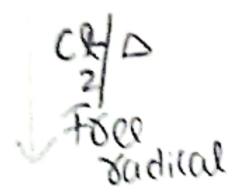
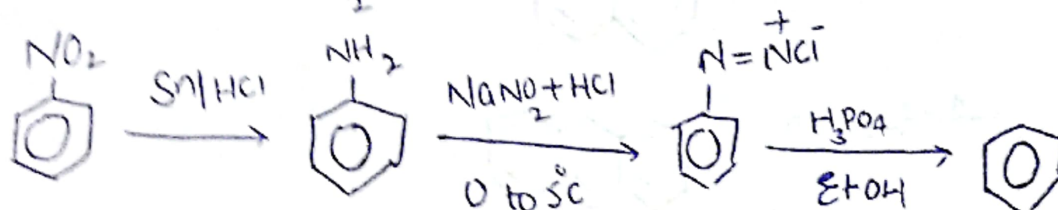
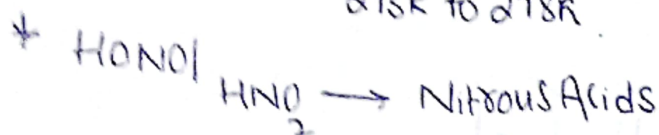
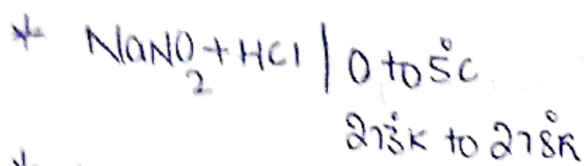
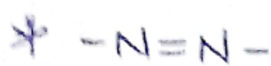
2) Cumene:



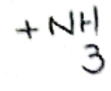
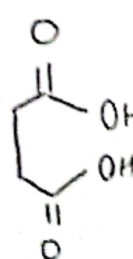
3) Glyoxal: CHOCHO.



Diazotisation :



OMSGAP (Oxalic, Malonic, Succinic,



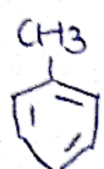
Glutaric, Adipic, Pimelic)

$\therefore$ allylic

1) Benzene :

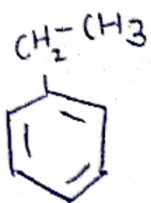


2)



Methyl Benzene (Toluene)

3)

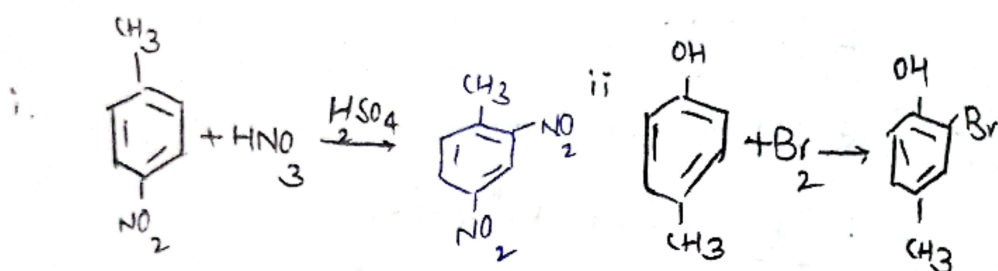
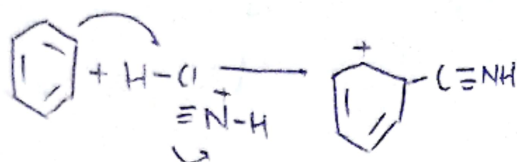
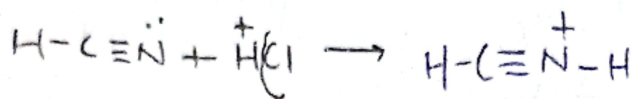


Ethyl Benzene.

Gattermann Rxn:

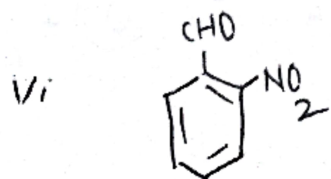
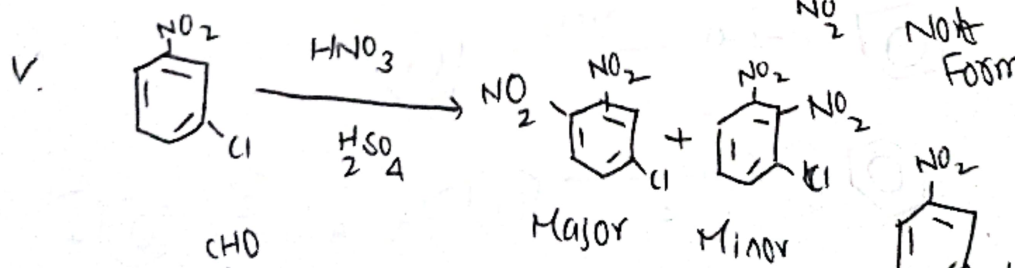
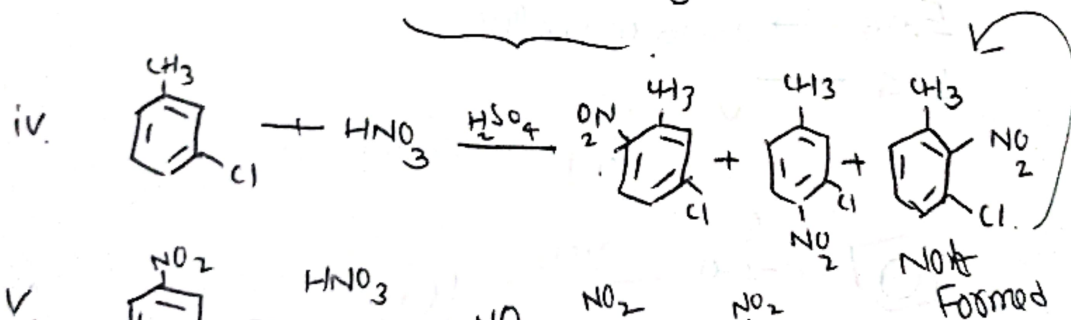
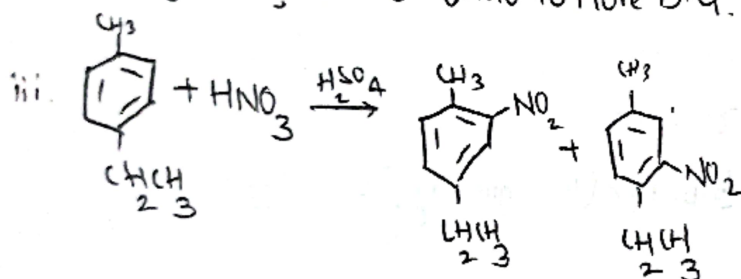
* Benzene $\rightarrow$ Benzaldehyde.

* $Zn(CN)_2 / HCl$ (2)
 HCl / HCN $\xrightarrow{H_2O}$



1,4-D₂O $\rightarrow$ E^+ Ortho to O.D.

1,4 $\rightarrow$ D₂O $\rightarrow$ E^+ Ortho to MORE D.G.



• if there is 2 more Directing
 the more deactivating group decides Orientation